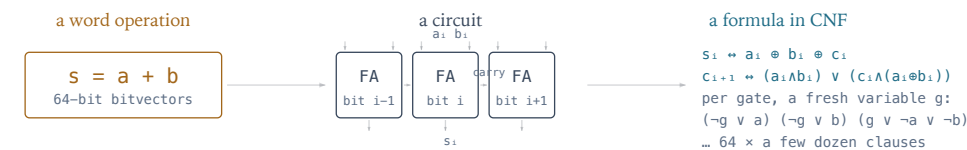


# Bounded Model Checking

SMT: bitvectors and arrays by bit-blasting. Bitme: the model of week 10 unrolled and handed to the solver of week 11 — and your own extension checked.

[← CC 11 · SAT](#)

# A SAT solver speaks bits. A program speaks words and memory. Bit-blasting connects them.



Every operation of the machine on words — add, compare, shift, load — is a circuit over the bits of its operands, and every gate is three or four clauses over the bits (Tseitin, 1968). An SMT formula over bitvectors becomes a SAT formula, much bigger, and goes to CDCL. This is the construction in Cook's proof that SAT is NP-hard, used as a tool.

arrays (memory) are added on demand:  $\text{read}(\text{write}(m, i, v), i) = v$ , and  $\text{read}(\text{write}(m, i, v), j) = \text{read}(m, j)$  for  $j \neq i$

Bitwuzla · Z3 · the solvers bitme drives

a word operation · a circuit · a formula

Satisfiability modulo theories. A propositional skeleton whose atoms are statements in a theory: bitvectors, with the machine's operations on fixed-width words; arrays, with read and write on an indexed store. SMT-LIB is the notation; Z3 and Bitwuzla read it, and BTOR2 too.

Bit-blasting. A 64-bit word is sixty-four variables; an addition is the adder circuit; every gate is three or four clauses with a fresh variable (Tseitin). Arrays get their axioms on demand. The SMT formula becomes a SAT formula, much bigger, and CDCL runs on it. Cook's construction, as engineering.

Bitme: unroll, ask, and read the **input** off the satisfying assignment.

```
$ ./rotor -c examples/symbolic/division-by-zero-3-35.c - 0  
$ tools/bitme.py -kmax 120 --use-bitwuzla examples/symbolic/division-by-zero-3-35-rotorized.btor2  
bitme bounded model checking: -kmin 0 -kmax 120  
0: initializing · 1: transitioning · ... · 75: transitioning  
76: core-0-division-by-zero  
76: vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv  
76: 150 state 71 input-buffer ; uninitialized input buffer  
76: state150-76 = (store ... #b0 #b00110000) ← byte 48, the character '0'  
89: core-0-division-by-zero ... #b00110010 ← byte 50, '2': the second division  
106: core-0-bad-exit-code ... #b00001100 ← byte 12: exit(0), the flagged exit
```

Bitme walks the steps and, at each, asks the solver about every bad state. Seventy-five steps of no while the program sets up its stack and calls malloc and read. Then, at step 76, where the division instruction is fetched, yes — and the assignment of the uninitialised input buffer is the failing input. Nobody told it to try '0'.

The step numbers move with the versions of rotor and bitme; check before class. The shape does not move: a bad state, a step, an input. And the same run on the students' own extensions is the assignment.

## WHAT A BOUND BUYS

A **no** up to  $k$  is a **theorem**.  
It says nothing about  $k + 1$ .

## TESTING · AND THE GRADER

One input per run. Shows presence. The grader's few hundred programs are a few hundred points in the space of week 2 of the introduction class.

## BOUNDED MODEL CHECKING

Every input, every path, up to  $k$  steps. Shows absence, within the bound, as a theorem. Pays in solver time that grows with  $k$ , and says nothing beyond it.

Rice forbade deciding whether a program divides by zero. This decides whether it does so within  $k$  steps — a question about a finite object. Choosing  $k$  is choosing what to give up: the sound-but-incomplete corner of week 9's table, with the incompleteness made explicit as a number.

Everything in the workshop around selfie is such a choice: rotor and bitme within a bound, the type checker for one property, buzzr complete about the crashes it happens to see. Each stated. Each applied to selfie itself.

## ASSIGNMENT

# rotor-check: verify your own extension.

- 1 A test program. In C\*, using your extension — the for loop, arrays, lazy evaluation, structs — with one input on which it misbehaves: an out-of-bounds index, a division by zero, a wrong exit code, reachable only for some input byte.
- 2 A property. State which of rotor's bad states the misbehaviour is, and the bound within which it is reachable.
- 3 The run. Compile the program with *your* selfie, generate its rotor model, run bitme with your bound. Submit the program, the property, the bound, and the input the solver found.
- 4 The check. The grader recompiles your program with your selfie, regenerates the model, runs bitme to your bound, and compares the verdict and the input.

**AND STRUCT-EXECUTION**

The other half of the capstone: field access  $p \rightarrow f$  as a load or store at the pointer plus the field's offset, types from the entry, the assertions on temporaries. Then use a struct in selfie.c — for the symbol table, if you dare — and self-compile.

Rules as always: selfie.c and the grammar; no warnings; self-compile passes. For rotor-check the submission is a directory with the program and a one-page note.

# Listen, then read.

## LISTEN

[Strauss · Elektra](#) — Solti, Nilsson, Vienna 1967. One act, no interruption, one question asked of every state: is it him? The recognition scene is the satisfying assignment found; the dance is what happens when the bound is reached.

## READ

[Handbook of Model Checking](#) — Clarke, Henzinger, Veith and Bloem, editors, 2018: where the field defines its terms; bounded model checking is one chapter. And, more technical, [A Tool for Checking ANSI-C Programs](#) — Clarke, Kroening and Lerda, 2004: CBMC, the bounded model checker for C that rotor and bitme are cousins of.